

Neal Daftary

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EDUCATION

Institute of Technology, Nirma University

Ahmedabad, India

B.Tech in Computer Science and Engineering (AI & ML)

Sep 2024 – Jun 2028 (Expected)

- JEE Mains 2024: **97.62 percentile** among more than 1.5 million candidates (Jan 2024)
- **Student Chairperson** (Sept 2025 - Present): Growing ACM ITNU to **150+ members**, evidenced by 2 flagship programs launched (Prompt to Prototype hackathon; a 4-track mentorship pipeline).

EXPERIENCE

Open Source Contributor

Jul 2026 – Present

Google DeepMind

Remote

- Contributed production code to **jax_privacy**, Google DeepMind's JAX-based differential privacy library, by authoring and merging pull requests that extended its DP feature set
- Reduced installation overhead for downstream users by refactoring the library's **chex** dependency to simplify setup requirements
- Earned direct validation from the maintainer team, with contributions reviewed by Mikhail Pravilov and approved by lead maintainer Ryan McKenna

Software Engineering Intern

Oct 2025 – Dec 2025

MZHubtech

Hamburg, Germany (Remote)

- Achieved **+57% performance** and **+40% SEO** gains (Core Web Vitals) by deploying an SSR **Next.js** app to **Azure App Service** via a **GitHub Actions** CI/CD pipeline
- Enabled scalable data persistence and automated transactional email for production traffic by integrating **Azure Cosmos DB** (NoSQL) and **SendGrid** SMTP into the application backend
- Shaped the product roadmap with actionable feasibility findings, adopted directly into planning discussions, by conducting R&D on agentic AI workflows and CX-automation chatbots

AI Intern

Jun 2025 – Sep 2025

8xSports

Bhilwara, India (Remote)

- Maintained **sub-800ms** retrieval latency at **50× scale (15,000 profiles)** by building a **FAISS IVF-PQ/OPQ** index with GPU batch encoding and nightly incremental updates
- Cut edge-device inference to **sub-20ms** by fine-tuning **YOLOv8s** (mosaic/mixup augmentation, IoU=0.35, cosine LR schedule) and exporting to **ONNX + TensorRT FP16**
- Improved athlete re-identification robustness under occlusion and lighting variance by fusing **DINOv2 ViT-S/14** patch embeddings with LBP texture histograms into multi-modal descriptors

PROJECTS

NavkaarAI: AI WhatsApp Agent Platform for Shopify D2C Brands | *Next.js, BullMQ, Redis, Supabase* | [GitHub](#) 2026

- Recovered abandoned carts and automated support at **20K+ messages/mo per merchant** by integrating Meta's WhatsApp Cloud API with Anthropic Claude (GPT-4o-mini fallback) over Shopify OAuth
- Sustained **>90% message delivery** on a horizontally scalable async backend by routing sends through **BullMQ** workers on Railway with Upstash Redis, and Supabase Postgres + Row-Level Security for tenant isolation
- Shipped a production **Turborepo** monorepo audited to **zero known bugs** across 10+ resolved issues by enforcing typed contracts, CI lint/typecheck, and Sentry observability pre-launch

mcptail: Zero-Config Observability Tool for MCP Servers | *TypeScript, Preact, Vite, Node* | [GitHub](#) | [npm](#) 2026

- Replaced manual **console.error** debugging with a single-command wiretap across Claude Code, Cursor, and VS Code, measured by **100% local** capture with no account or cloud dependency, by engineering a transparent MCP proxy and shipping it to npm
- Exposed per-tool token cost and **p50/p95 latency** for every session with zero disruption to live agents by building a live SSE dashboard with searchable timeline, error filter, and one-click replay
- Enforced a merge bar of **57 automated tests** across 9 suites with lint and dual type-checks on every commit by wiring Biome, tsc, and Vitest into a pre-commit hook and CI

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL

Backend: FastAPI, Next.js, Node.js, Docker, Redis (BullMQ), PostgreSQL, REST/SSE

ML/CV: PyTorch, ONNX, TensorRT, FAISS, YOLOv8, SegFormer, DINOv2, SAM2, OpenCV, HuggingFace

LLM/MLOps: LangChain, LangGraph, LangSmith, Prompt Engineering, MLflow, DVC, Airflow, GitHub Actions, Grafana

Cloud/Tools: Azure (App Service, Cosmos DB), AWS (S3, ECR, ECS, SageMaker), Vercel, Railway, Google Colab, Claude Code

AWARDS & ACHIEVEMENTS

CatBoost-Driven Anomaly Detection in Industrial Robotic Arms (IEEE Xplore): Published in *IEEE Sensor Letters* (SCI Q3, IF 2.2), Jan 2026. Achieved **97.20% precision**, **F1 0.9718** on 200K+ samples (CASPER dataset), outperforming standard baselines

4th National Rank, 6th Mitsubishi Electric Cup 2026: Improved paint-defect segmentation accuracy from 55% to **92.35%** (86% dimensional precision) by developing SpectraScan, a DINOv2 + FPN-UNet system ([GitHub](#))

Track Winner & Top-5 Finalist, HACKaMINeD 2026 (2,200+ participants): Built LUMIN.AI, explainable ML for solar-inverter predictive maintenance ([GitHub](#))

Letter of Recommendation, 8xSports: Awarded for high-impact execution on a production jersey re-identification system

Complete MLOps Bootcamp (Udemy)-Certificate

Advanced Learning Algorithms, DeepLearning.AI/Stanford (Coursera): **99.77%** - Certificate